

relatorio

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Imports

```
#install.packages("plotly")  
library("plotly")  
  
source("functions.R")
```

Função Gaussiana

```
$$  
f(x, y) = K \cdot e^{-c(x^2 + y^2)}  
$$
```

$$f(x, y) = k \cdot \exp[-c(x^2 + y^2)]; k, c > 0$$

k : Amplitude máxima da função; $f(0,0) = k$

c: Abertura e decaimento da curva;

```
f <- function(x,y, k=3, c=2){  
  
  return( k*exp(-c(x^2 + y^2)))  
}
```

Objetivo

Usar polinômios de duas variáveis para aproximar a função gaussiana;

Geração de pontos aleatórios sobre o retângulo $[-4,4] \times [-4,4]$

```
set.seed(28)  
  
# número de pontos  
n<- 1000  
  
min_<- -4  
max_<- 4  
  
x<- runif(n, min= min_, max= max_)  
y<- runif(n, min= min_, max= max_)  
  
z<- f(x,y)
```

```

pontos<- cbind(x,y, z)

matriz<- data.frame(pontos)

print(matriz[1:5,])

##           x           y           z
## 1 -3.771398  1.5365098 1.882134e-07
## 2 -3.293045 -1.4822836 6.506570e-06
## 3 -0.205058 -0.7228664 1.705789e+00
## 4  3.083021 -2.2774995 1.248696e-06
## 5 -3.267464 -0.9786600 2.657651e-05

plot_pontos_3d(x,y,z)

```

Possível problema

Há muitos pontos na região em que $f(x,y)$ é próximo de zero ...

Modelo 1 [$m_1(x,y) = b_0 + b_1x + b_2y$]

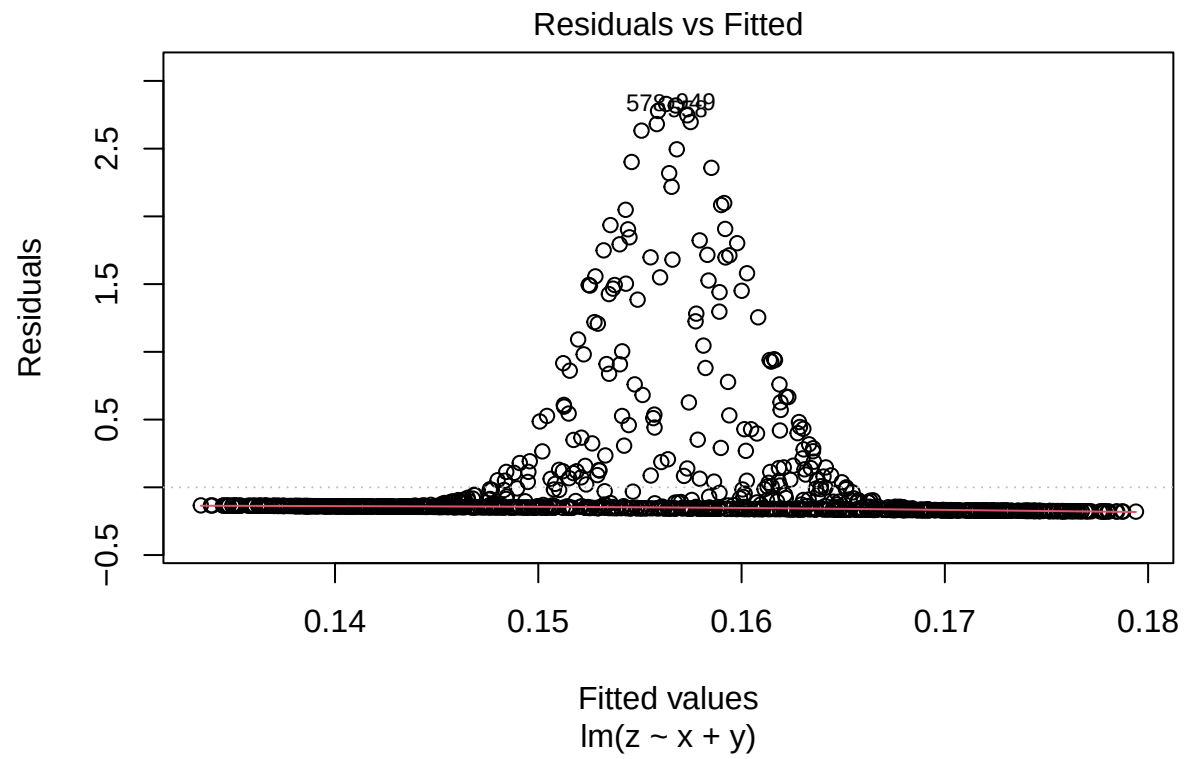
```

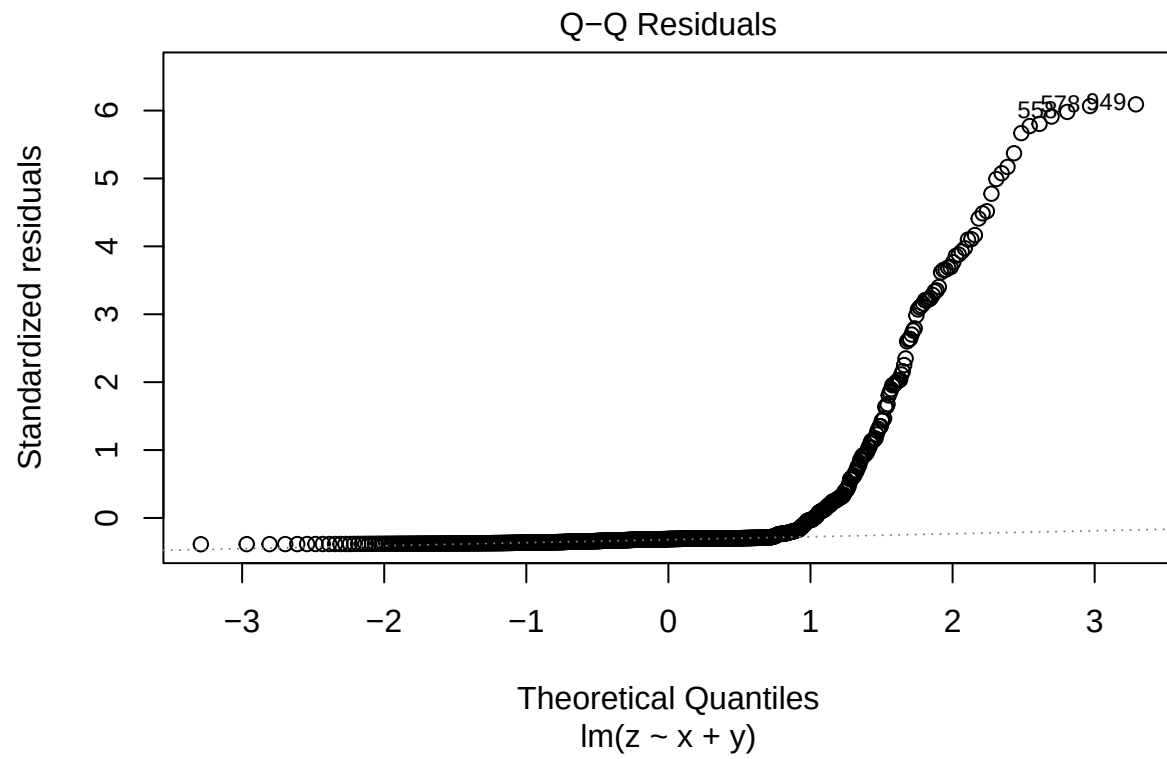
modelo1<- lm(z ~ x + y, data=matriz)
summary(modelo1)

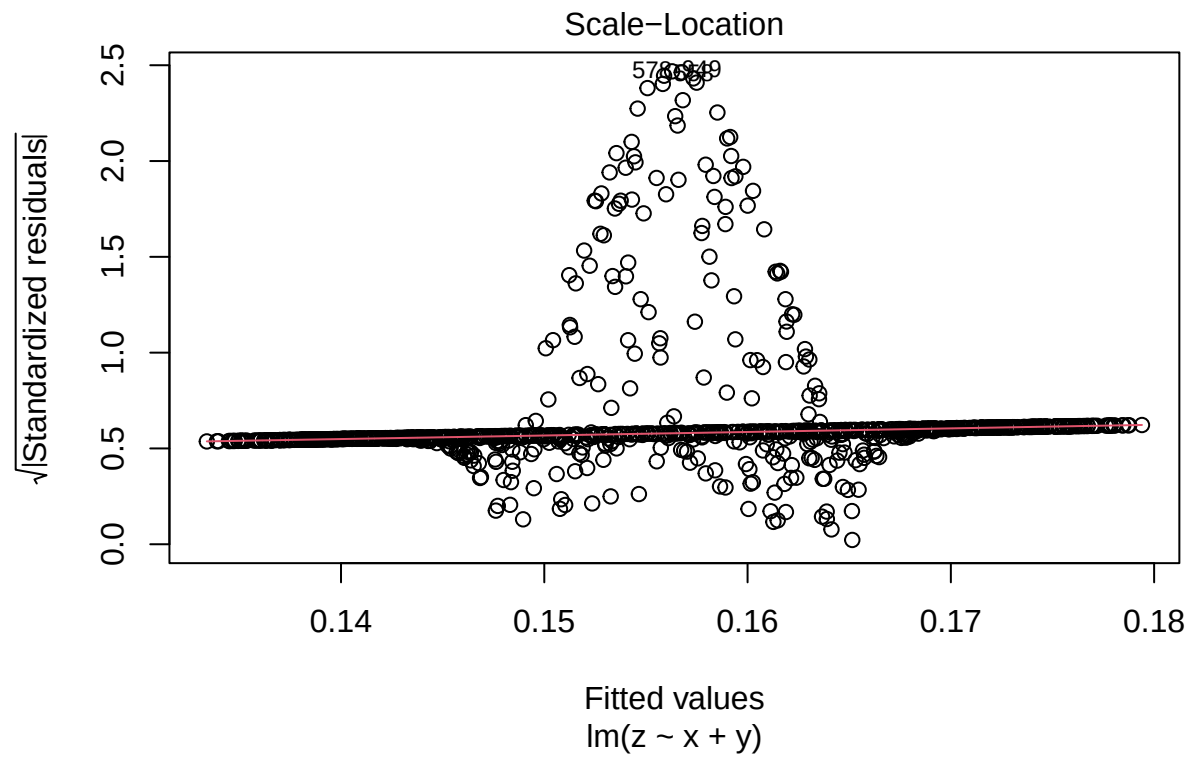
##
## Call:
## lm(formula = z ~ x + y, data = matriz)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.1794 -0.1624 -0.1442 -0.1349  2.8299
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.1564746  0.0147319  10.621  <2e-16 ***
## x            0.0051034  0.0063594   0.802   0.422
## y           -0.0007783  0.0064509  -0.121   0.904
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4648 on 997 degrees of freedom
## Multiple R-squared:  0.0006603, Adjusted R-squared:  -0.001344
## F-statistic: 0.3294 on 2 and 997 DF,  p-value: 0.7195

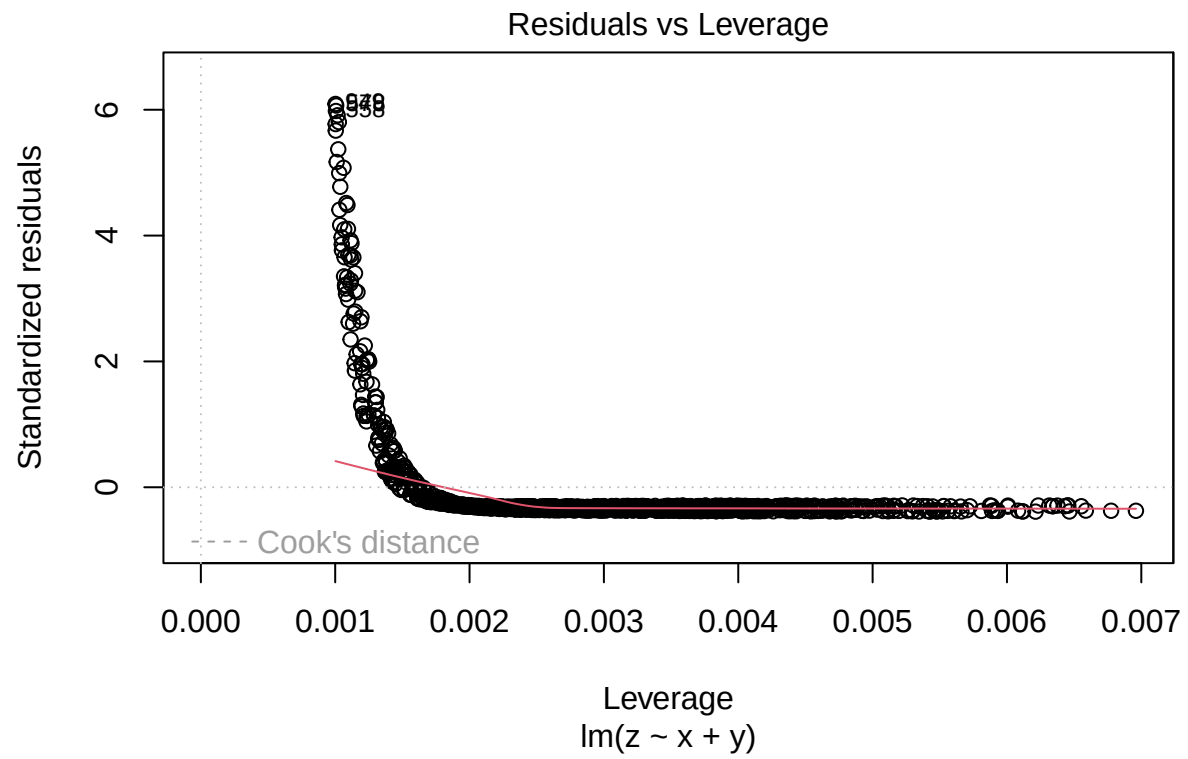
plot(modelo1)

```

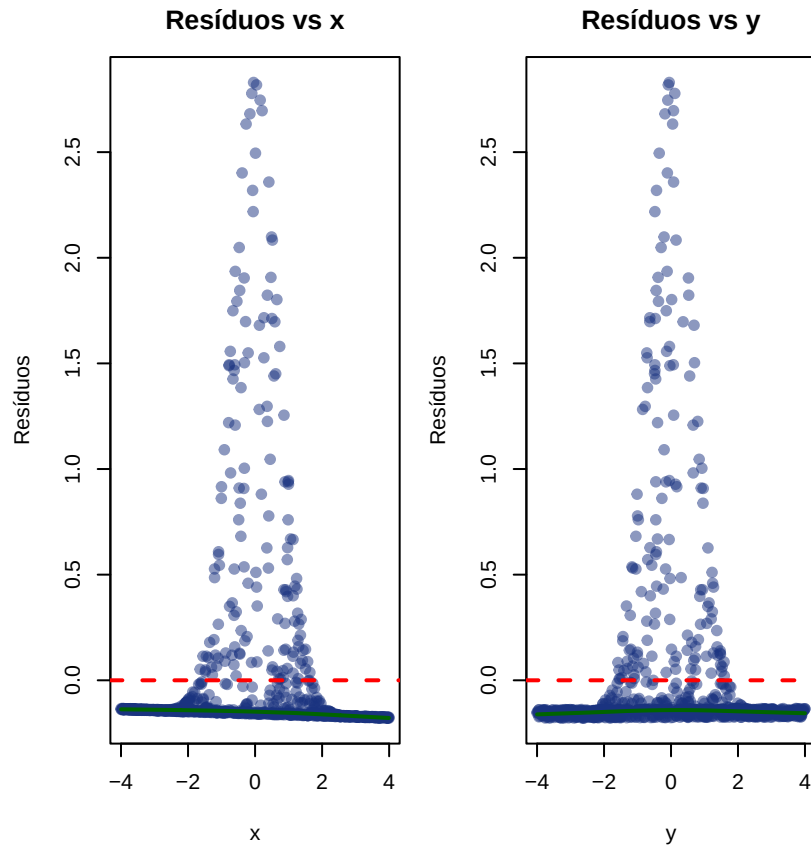








```
plot_residuos_preditores(modelo1, matriz)
```



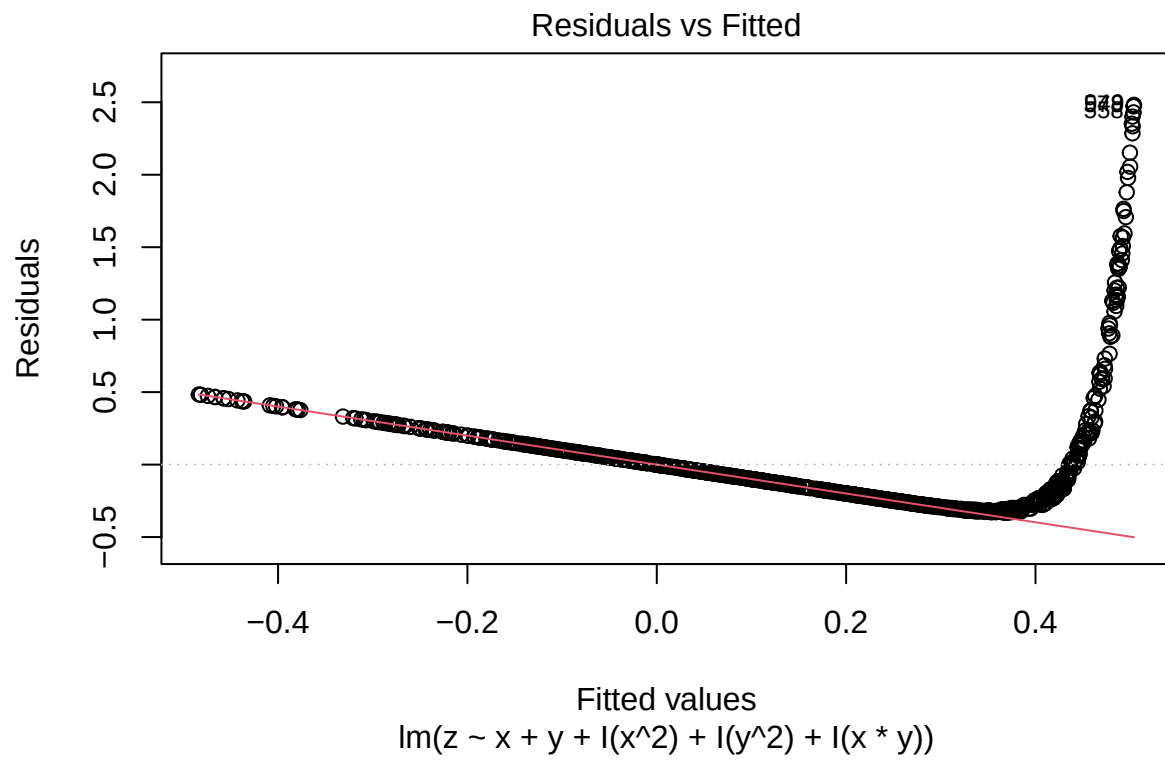
Modelo 2 [$m_2(x,y) = b_0 + b_1 \cdot x + b_2 \cdot y + b_3 \cdot x^2 + b_4 \cdot y^2 + b_5 \cdot x \cdot y$]

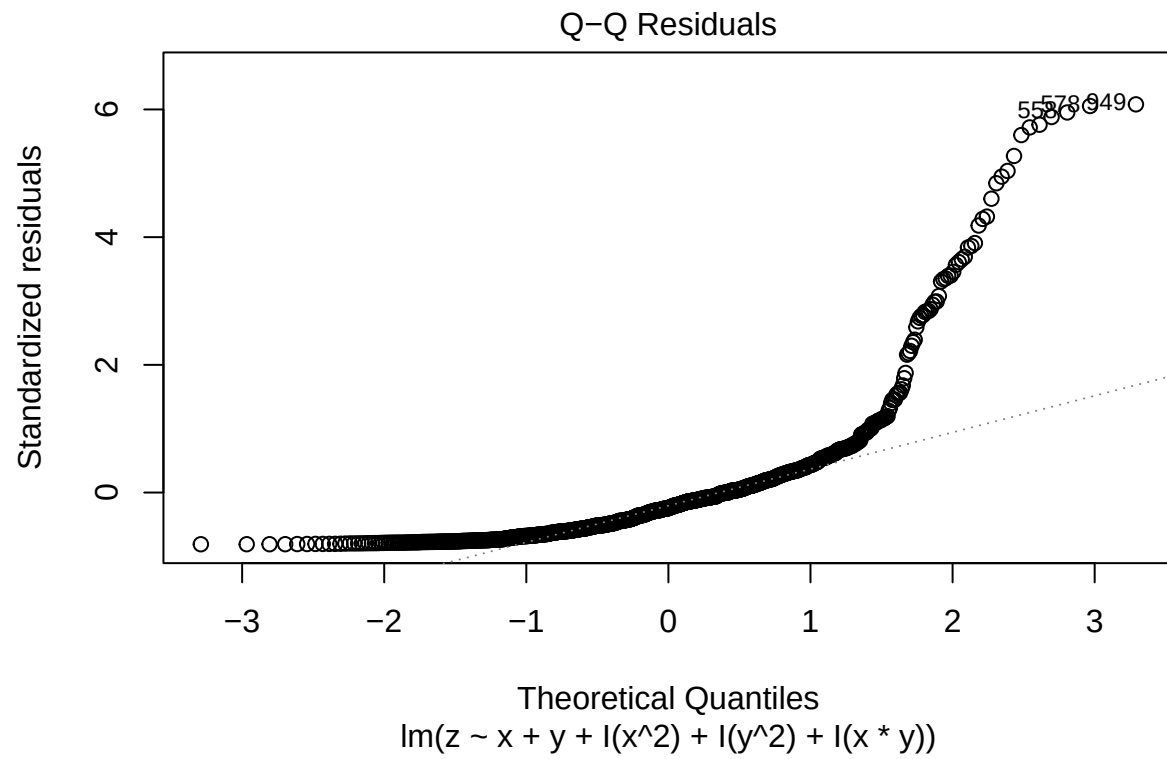
```
modelo2<- lm(z ~ x + y + I(x^2) + I(y^2) + I(x*y), data=matriz)
summary(modelo2)
```

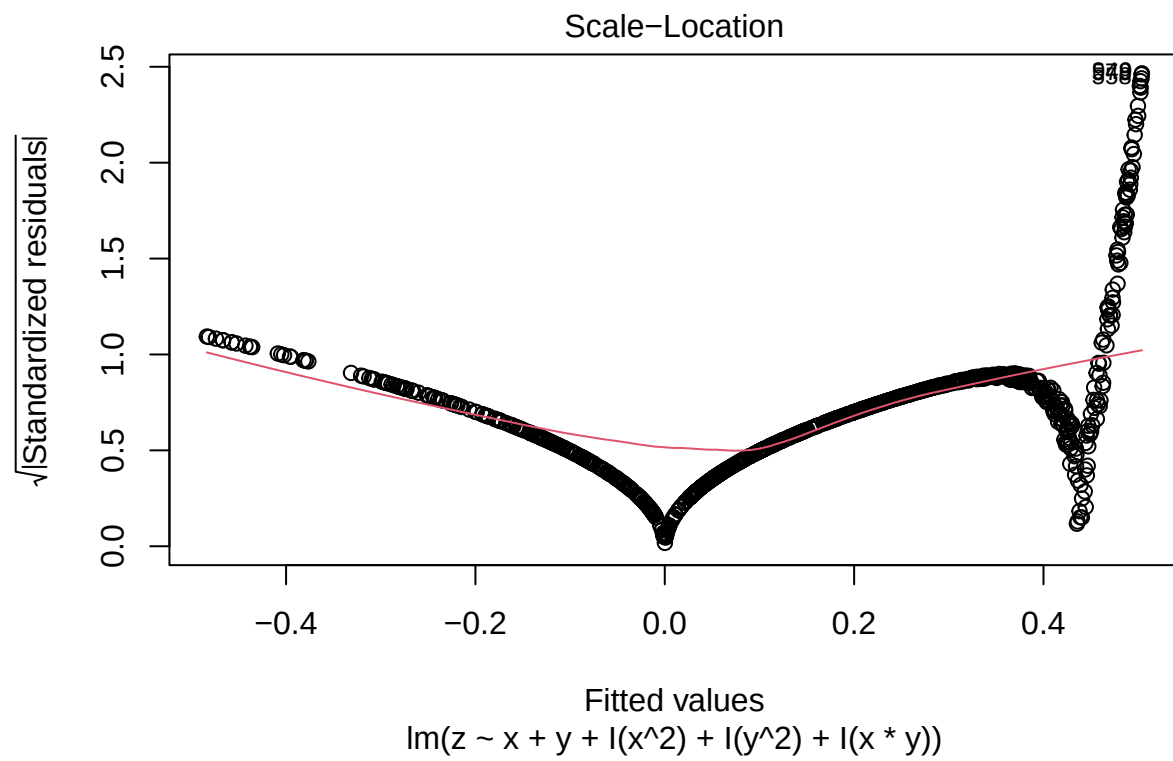
```
##
## Call:
## lm(formula = z ~ x + y + I(x^2) + I(y^2) + I(x * y), data = matriz)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.33061 -0.24182 -0.09896  0.07386  2.48216
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.5040320  0.0240662  20.944  <2e-16 ***
## x           -0.0029806  0.0056175  -0.531   0.596
## y            0.0002422  0.0057062   0.042   0.966
## I(x^2)       -0.0326258  0.0027186 -12.001  <2e-16 ***
## I(y^2)       -0.0334508  0.0027811 -12.028  <2e-16 ***
## I(x * y)      0.0016940  0.0024453   0.693   0.489
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

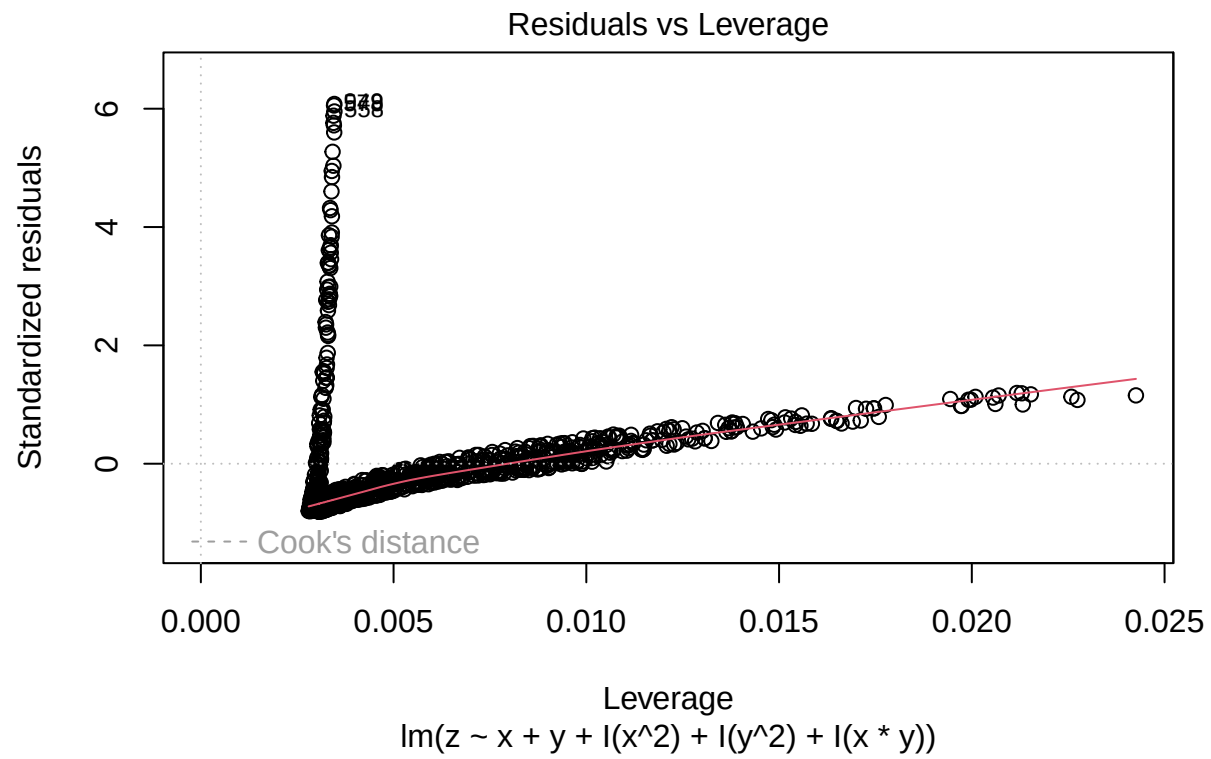
```
##
## Residual standard error: 0.4089 on 994 degrees of freedom
## Multiple R-squared:  0.2287, Adjusted R-squared:  0.2248
## F-statistic: 58.93 on 5 and 994 DF,  p-value: < 2.2e-16
```

```
plot(modelo2)
```

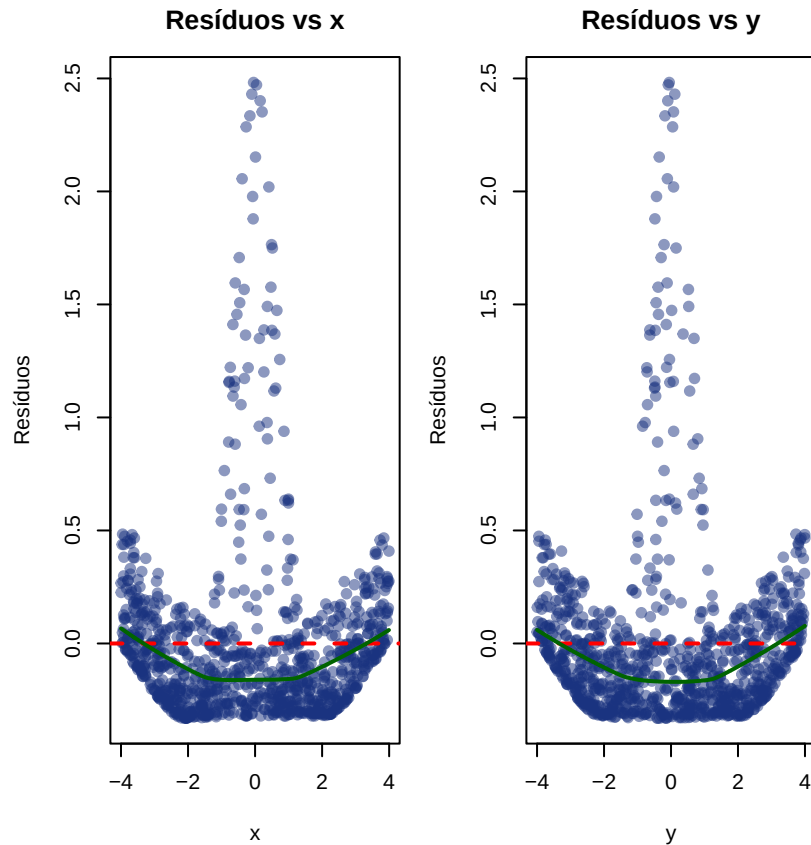








```
plot_residuos_preditores(modelo2, matriz)
```

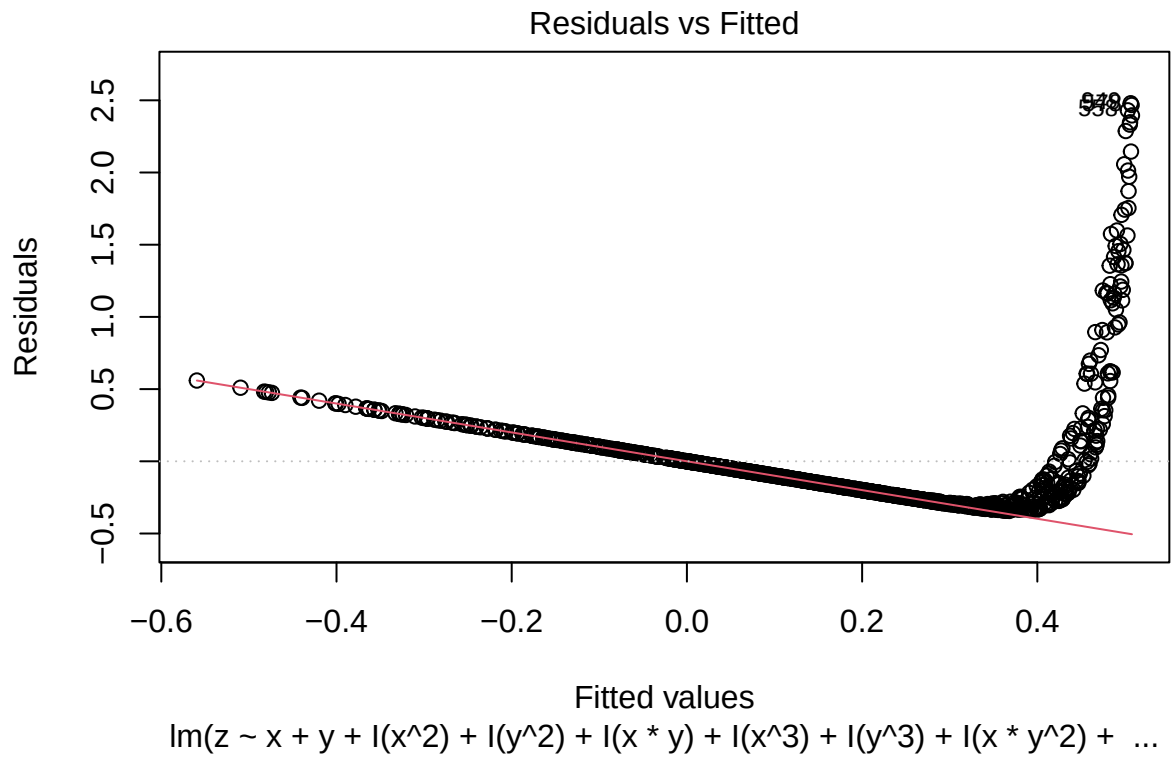


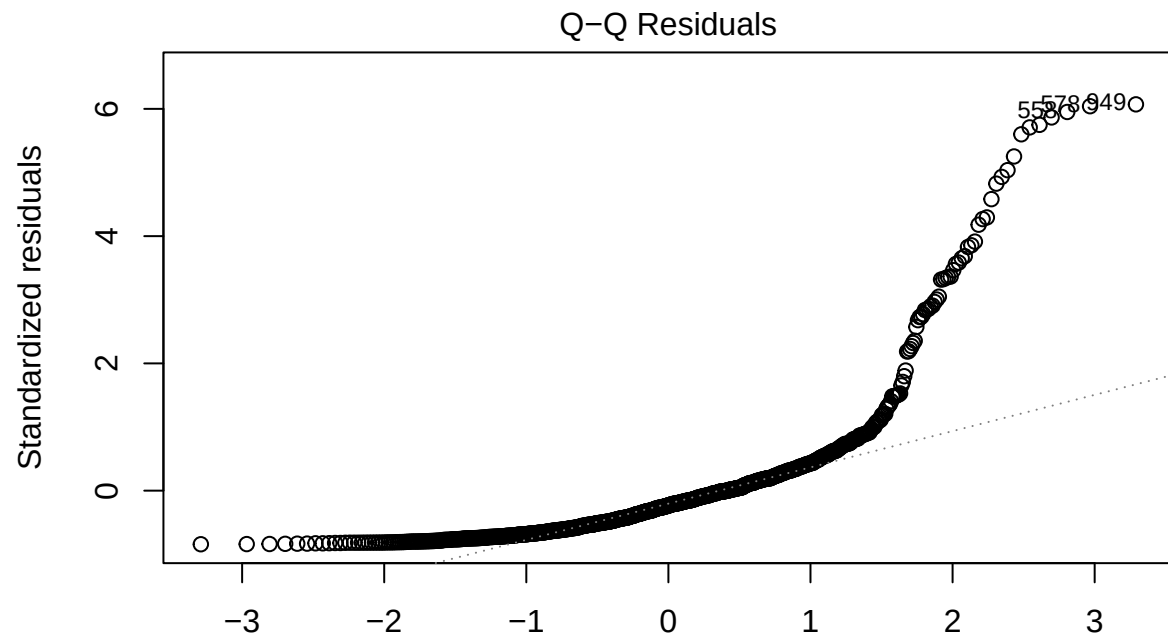
Modelo 3 [$m_2(x,y) = b_0 + b_1 \cdot x + b_2 \cdot y + b_3 \cdot x^2 + b_4 \cdot y^2 + b_5 \cdot x \cdot y + b_6 \cdot x^3 + b_7 \cdot y^3$]

```
modelo3<- lm(z ~ x + y + I(x^2) + I(y^2) + I(x*y) + I(x^3) + I(y^3) + I(x*y^2) + I(y*x^2), data=matriz)
summary(modelo3)
```

```
##
## Call:
## lm(formula = z ~ x + y + I(x^2) + I(y^2) + I(x * y) + I(x^3) +
##      I(y^3) + I(x * y^2) + I(y * x^2), data = matriz)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.34361 -0.23992 -0.09491  0.07312  2.47957
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.065e-01  2.413e-02  20.987  <2e-16 ***
## x             9.202e-03  1.505e-02   0.611   0.541
## y            -1.301e-02  1.549e-02  -0.840   0.401
## I(x^2)        -3.277e-02  2.726e-03 -12.023  <2e-16 ***
## I(y^2)        -3.398e-02  2.804e-03 -12.119  <2e-16 ***
## I(x * y)       1.900e-03  2.465e-03   0.771   0.441
## I(x^3)        -3.696e-04  1.330e-03  -0.278   0.781
```

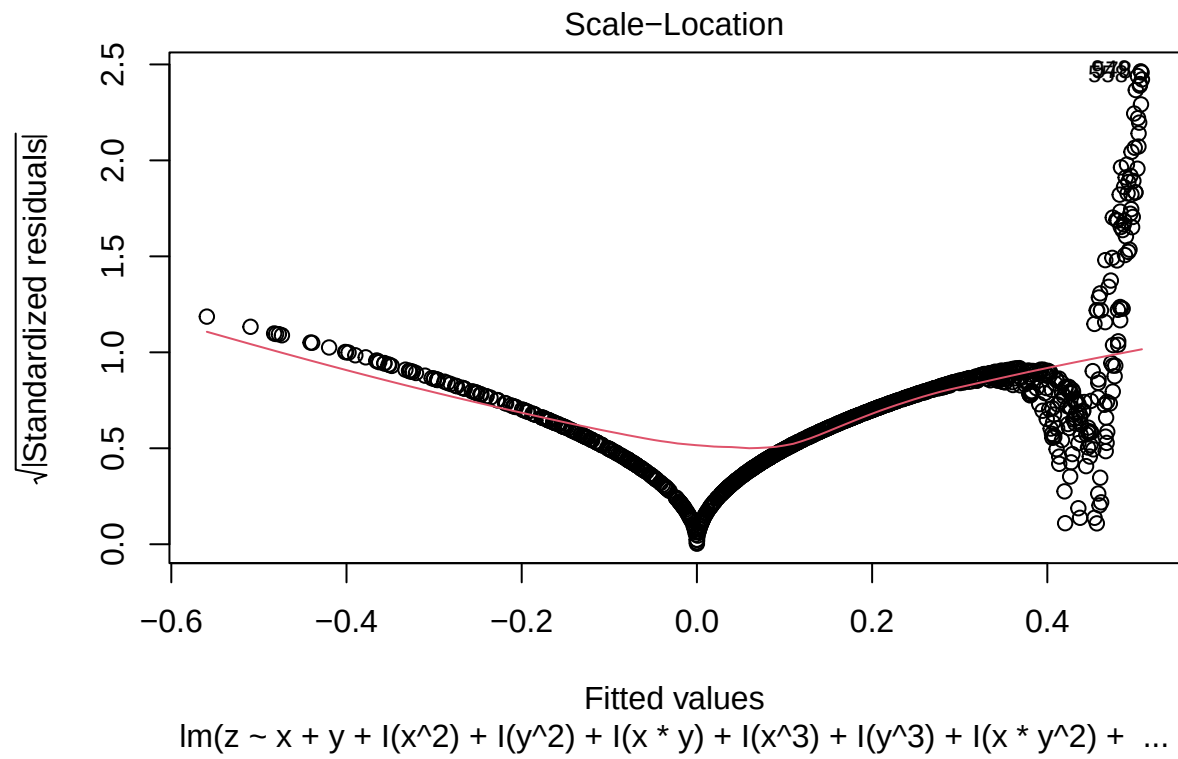
```
## I(y^3)      1.444e-03  1.364e-03  1.058  0.290
## I(x * y^2) -1.712e-03  1.229e-03 -1.393  0.164
## I(y * x^2)  3.144e-05  1.173e-03  0.027  0.979
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4091 on 990 degrees of freedom
## Multiple R-squared:  0.2311, Adjusted R-squared:  0.2241
## F-statistic: 33.06 on 9 and 990 DF,  p-value: < 2.2e-16
plot(modelo3)
```

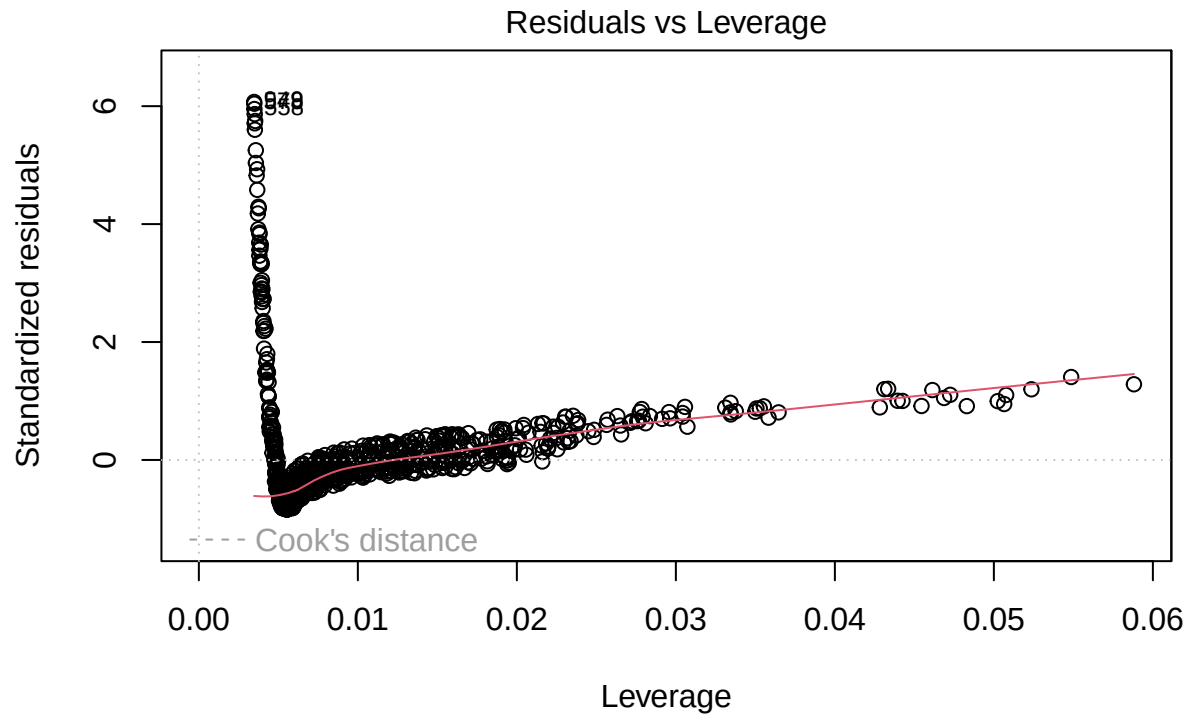




Theoretical Quantiles

$\text{Im}(z \sim x + y + I(x^2) + I(y^2) + I(x * y) + I(x^3) + I(y^3) + I(x * y^2) + \dots$





$\text{lm}(z \sim x + y + \text{I}(x^2) + \text{I}(y^2) + \text{I}(x * y) + \text{I}(x^3) + \text{I}(y^3) + \text{I}(x * y^2) + \dots$

```
plot_residuos_preditores(modelo3, matriz)
```